

# LEARN Checkpoint 1 Practice Set

100 items • Five standards • Both segments

Built by an Indiana middle school math teacher, aligned to the 2023 Indiana Academic Standards.

|                                          |                                                                                                                                                                                                                                          |
|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CHECKPOINT 1 IS THE SEGMENTED ONE</b> | Segment 1, no calculator: 8.NS.2 • 8.NS.3. Segment 2, calculator permitted: 8.NS.4 • 8.AF.1 • 8.AF.2. Students meet both segments in one sitting, and this set is ordered the same way.                                                  |
| <b>WHY THAT MATTERS</b>                  | Grade 6 splits Checkpoint 2 and grade 7 splits Checkpoint 3. Grade 8 splits Checkpoint 1. Practice that ignores the split does not simulate the test.                                                                                    |
| <b>REPORTING CATEGORIES</b>              | Number Sense (8.NS.2, 8.NS.3, 8.NS.4) • Algebra and Functions (8.AF.1, 8.AF.2)                                                                                                                                                           |
| <b>WORTH KNOWING</b>                     | 8.AF.2 appears on Checkpoint 1 and nowhere else — not on Checkpoint 2, Checkpoint 3, or the summative. It is the only grade 8 standard assessed exactly once all year, and this is the only window in which it is taught to a live test. |

Every item is labeled by proficiency tier.

## Approaching Proficiency 25

Recognize and identify. The on-ramp — students name the right idea before producing it.

## At Proficiency 40

Produce. Students carry out the skill in a real-world or multi-step setting.

## Above Proficiency 35

Justify or generalize. Students verify a claim, find a missing exponent or constant, or explain why a rule has to hold.

No Below Proficiency items; 75% of the set sits in the top two tiers. Two standards, 8.AF.1 and 8.AF.2, have no published difficulty ladder in their specifications — their bands here are derived from the stated content limits and labeled as such on the standard page.

Original items. Built to the published Indiana Academic Standards and the Grade 8 assessment structure; not affiliated with or endorsed by the Indiana Department of Education.

# SEGMENT 1 — NO CALCULATOR

## Real Numbers

Irrational approximations and the properties of exponents. Students meet these two standards before any calculator is handed out.

### 8.NS.2

Use rational approximations of irrational numbers to compare the size of irrational numbers, plot them approximately on a number line, and estimate the value of expressions involving irrational numbers.

| REPORTING CATEGORY | SUBDOMAIN    | CALCULATOR    | ITEMS |
|--------------------|--------------|---------------|-------|
| Number Sense       | Real Numbers | Not permitted | 20    |

#### ACADEMIC VOCABULARY

rational numbers, irrational numbers, non-perfect square, perfect square, estimate

#### HOW DIFFICULTY ESCALATES ON THIS STANDARD

Easy:  $\sqrt{2}$ ,  $\sqrt{3}$ , and  $\pi$ ; number lines marked in integer intervals. Medium: square roots of prime numbers less than 100; number lines marked in halves. Difficult: square roots of composite numbers, expressions,  $k\pi$ , or values greater than 100; number lines marked in quarters.

### Approaching Proficiency 5 items — Recognize the idea and name it.

#### 1 APPROACHING PROFICIENCY 8.NS.2 DOK 1 EASY

Between which two consecutive integers does  $\sqrt{2}$  lie?

- A. 1 and 2
- B. 3 and 4
- C. 0 and 1
- D. 2 and 3

#### 2 APPROACHING PROFICIENCY 8.NS.2 DOK 1 EASY

Between which two consecutive integers does  $\sqrt{3}$  lie?  
Explain using perfect squares.

#### 3 APPROACHING PROFICIENCY 8.NS.2 DOK 1 EASY

Which integer is the best approximation of  $\pi$ ?

- A. 2
- B. 3
- C. 4
- D. 5

#### 4 APPROACHING PROFICIENCY 8.NS.2 DOK 1 MEDIUM

Between which two consecutive integers does  $\sqrt{53}$  lie?

#### 5 APPROACHING PROFICIENCY 8.NS.2 DOK 1 MEDIUM

A number line is marked in halves. Which point is closest to  $\sqrt{31}$ ?

- A. 5
- B. 5.5
- C. 6
- D. 6.5

## At Proficiency 8 items — Produce the result in context.

**6** AT PROFICIENCY 8.NS.2 DOK 2 EASY

Which inequality is true?

- A.  $\pi < 3$
- B.  $\sqrt{3} < 1$
- C.  $\sqrt{2} < \pi$
- D.  $\sqrt{2} > 2$

**7** AT PROFICIENCY 8.NS.2 DOK 2 EASY

Write an inequality comparing  $\sqrt{3}$  and 1.5. Justify it with perfect squares.

**8** AT PROFICIENCY 8.NS.2 DOK 2 MEDIUM

Order these values from least to greatest:  $\sqrt{47}$ , 6.5,  $\sqrt{41}$ .

**9** AT PROFICIENCY 8.NS.2 DOK 2 MEDIUM

Which value is greater,  $\sqrt{71}$  or 8.5?

- A.  $\sqrt{71}$ , because  $8^2 = 64$
- B.  $\sqrt{71}$ , because  $71 > 8.5$
- C. They are equal.
- D. 8.5, because  $8.5^2 = 72.25$  and  $72.25 > 71$

**10** AT PROFICIENCY 8.NS.2 DOK 2 MEDIUM

A number line runs from 7 to 9 marked in halves. Between which two marks does  $\sqrt{61}$  fall, and which mark is it nearer?

**11** AT PROFICIENCY 8.NS.2 DOK 2 DIFFICULT

Write an inequality comparing  $2\pi$  and 6.5. Justify your answer.

**12** AT PROFICIENCY 8.NS.2 DOK 2 DIFFICULT

Which value is closest to  $\sqrt{150}$ ?

- A. 11.5
- B. 12.25
- C. 13
- D. 75

**13** AT PROFICIENCY 8.NS.2 DOK 2 DIFFICULT

Order from least to greatest:  $3\pi$ ,  $\sqrt{90}$ , 9.5.

## Above Proficiency 7 items — Justify the reasoning, or work the skill backward.

**14** ABOVE PROFICIENCY 8.NS.2 DOK 3 MEDIUM

Which expression has the greater value,  $2\pi$  or 6.5?

- A. 6.5, because  $2\pi \approx 6.28$  and  $6.28 < 6.5$
- B. They are equal.
- C.  $2\pi$ , because  $2 \times 3.5 = 7$
- D.  $2\pi$ , because  $\pi$  is greater than 3

**18** ABOVE PROFICIENCY 8.NS.2 DOK 3 DIFFICULT

Order from least to greatest:  $3\pi$ ,  $\sqrt{81}$ , 9.5.

## Above Proficiency 7 items — Justify the reasoning, or work the skill backward.

**15** ABOVE PROFICIENCY 8.NS.2 DOK 3 MEDIUM

A student writes  $\pi + 3 = 6$  and says the sum is exact. Explain what is wrong, and give a better statement about the value of  $\pi + 3$ .

---

---

**16** ABOVE PROFICIENCY 8.NS.2 DOK 3 MEDIUM

Explain how you know  $4\pi$  is greater than 12 without finding a decimal value for  $4\pi$ .

---

---

**17** ABOVE PROFICIENCY 8.NS.2 DOK 3 DIFFICULT

Which expression has the **greatest** value?

- A. 7.5
  - B.  $2\pi + 1$
  - C.  $\sqrt{64}$
  - D.  $3\pi - 2$
- 

**19** ABOVE PROFICIENCY 8.NS.2 DOK 3 DIFFICULT

Name an integer that lies between  $2\pi$  and  $3\pi$ , and justify your choice.

---

---

**20** ABOVE PROFICIENCY 8.NS.2 DOK 3 DIFFICULT

A student claims that  $\sqrt{49} + \pi$  must be rational “because  $\sqrt{49}$  is a whole number.” Determine whether the sum is rational or irrational, and explain what the student overlooked.

---

---